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FINAL YEAR PROJECT PROPOSAL

JobRix

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Domain Overview

The Technology Landscape

- Artificial Intelligence (AI)
- Natural Language Processing (NLP)
- Computer Vision (OpenCV)
- Machine Learning

Core Project Focus

The objective of JobRix is to develop a unified recruitment platform that supports both candidates and recruiters. The system evaluates resumes, recommends suitable job opportunities, conducts AI-based interviews, generates detailed performance reports, and provides actionable insights to improve hiring decisions and candidate preparation.

Context

Current recruitment platforms primarily focus on job discovery and application management, while resume evaluation, interview preparation, candidate assessment, and hiring analytics are often handled using separate tools. This fragmented workflow increases manual effort, limits personalized feedback, and reduces the overall efficiency of the recruitment process.

Problem Statement

The Gap

Current recruitment platforms primarily focus on job listings and application management.

Resume evaluation, interview preparation, and career guidance are often handled through separate tools.

Recruiters rely on multiple systems for resume screening, interview evaluation, and hiring analytics.

This fragmented workflow increases manual effort and creates inconsistencies in candidate assessment.

Limited integration reduces the overall efficiency of the recruitment process.

Impact

The lack of an integrated recruitment workflow results in longer hiring cycles, repetitive manual effort, and limited feedback for candidates. Recruiters spend significant time switching between multiple tools, while candidates have limited visibility into their performance and areas for improvement, affecting both hiring quality and candidate experience.

In Clear Words...

Develop JobRix, an AI-assisted recruitment platform that unifies resume analysis, intelligent job matching, AI-based interviews, candidate performance evaluation, and recruiter analytics into a single system, enabling a more efficient, transparent, and data-driven hiring process.

Objectives & Project Scope

Primary Objectives

Enable students to bridge the gap between academic learning and industry expectations by identifying skill gaps and providing personalized recommendations for improvement.

Provide AI-assisted resume evaluation, mock interviews, and performance feedback to help students improve their technical, communication, and interview skills before applying for jobs.

Develop a unified platform that integrates ATS-based resume analysis, intelligent job matching, AI-assisted interviews, and candidate evaluation to support efficient interactions between job seekers and recruiters.

Out of Scope

- Final hiring decisions and candidate selection.
- Employee onboarding and HR management.
- Payroll, attendance, and employee lifecycle management.
- Integration with third-party recruitment (e.g., LinkedIn, Naukri, Indeed).
- Background verification and document validation.



Technical Scope - In Scope

- Develop a simple, responsive, and user-friendly interface for both job seekers and recruiters.
- Evaluate resumes using an ATS-based scoring system and identify areas for improvement.
- Conduct AI-assisted mock interviews and generate personalized performance reports.
- Recommend suitable job opportunities and learning resources based on candidate skills and performance.
- Provide recruiters with candidate insights, interview reports, and analytics to support the evaluation process.
- Integrate resume analysis, job matching, AI interviews, and recruiter management into a unified recruitment platform.

DESIGN PROCESS

A user-centric approach to build intuitive and efficient recruitment experience



Literature Survey / Prior Work

Analysis of existing platforms in the recruitment space and the gaps that JobRix addresses.

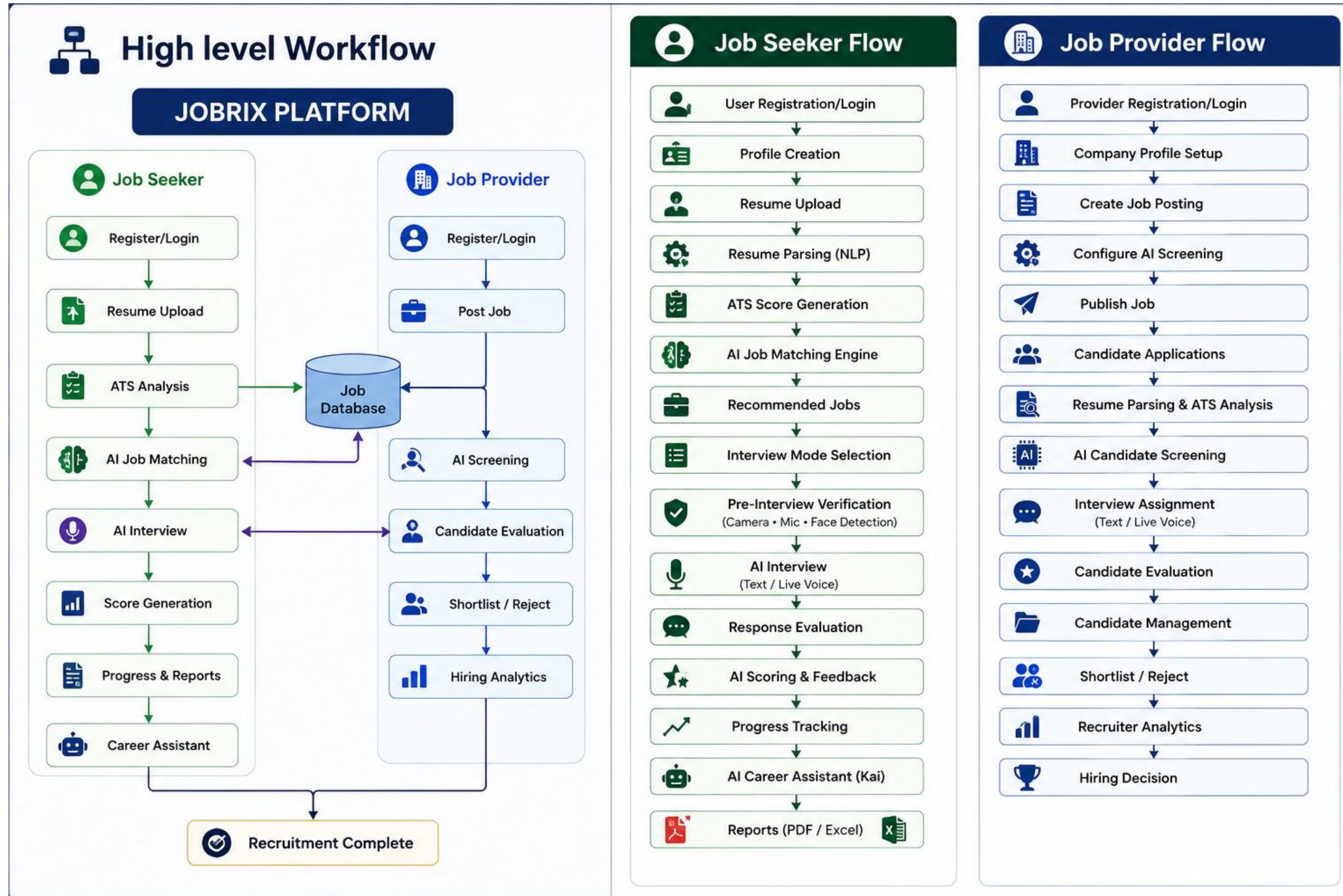
Existing Platform	Solution Type	Strengths	Limitations	Gap Addressed by JobRix
 LinkedIn	Professional Networking & Job Portal	✓ Large professional network, job search, and recruiter connectivity.	✗ Limited interview preparation, no ATS feedback, no integrated skill-gap analysis.	✓ Integrates resume analysis, AI interviews, job recommendations, and personalized performance feedback in one platform.
 Indeed	Job Search & Recruitment Platform	✓ Extensive job listings, resume hosting, and AI-assisted job recommendations.	✗ Focuses mainly on job applications; limited interview preparation and learning guidance.	✓ Provides ATS analysis, AI mock interviews, learning recommendations, and recruiter analytics together.
 Jobscan	ATS Resume Optimization	✓ ATS compatibility analysis and keyword optimization.	✗ Limited to resume optimization; no interview assessment, job matching, or recruiter workflow.	✓ Extends beyond ATS scoring by integrating interviews, job matching, and performance tracking.
 Interviewing.io	Technical Mock Interview Platform	✓ Realistic technical interview practice with detailed feedback.	✗ Primarily interview-focused; lacks resume evaluation, job recommendations, and recruiter modules.	✓ Combines interview practice with ATS analysis, career guidance, and recruiter support.



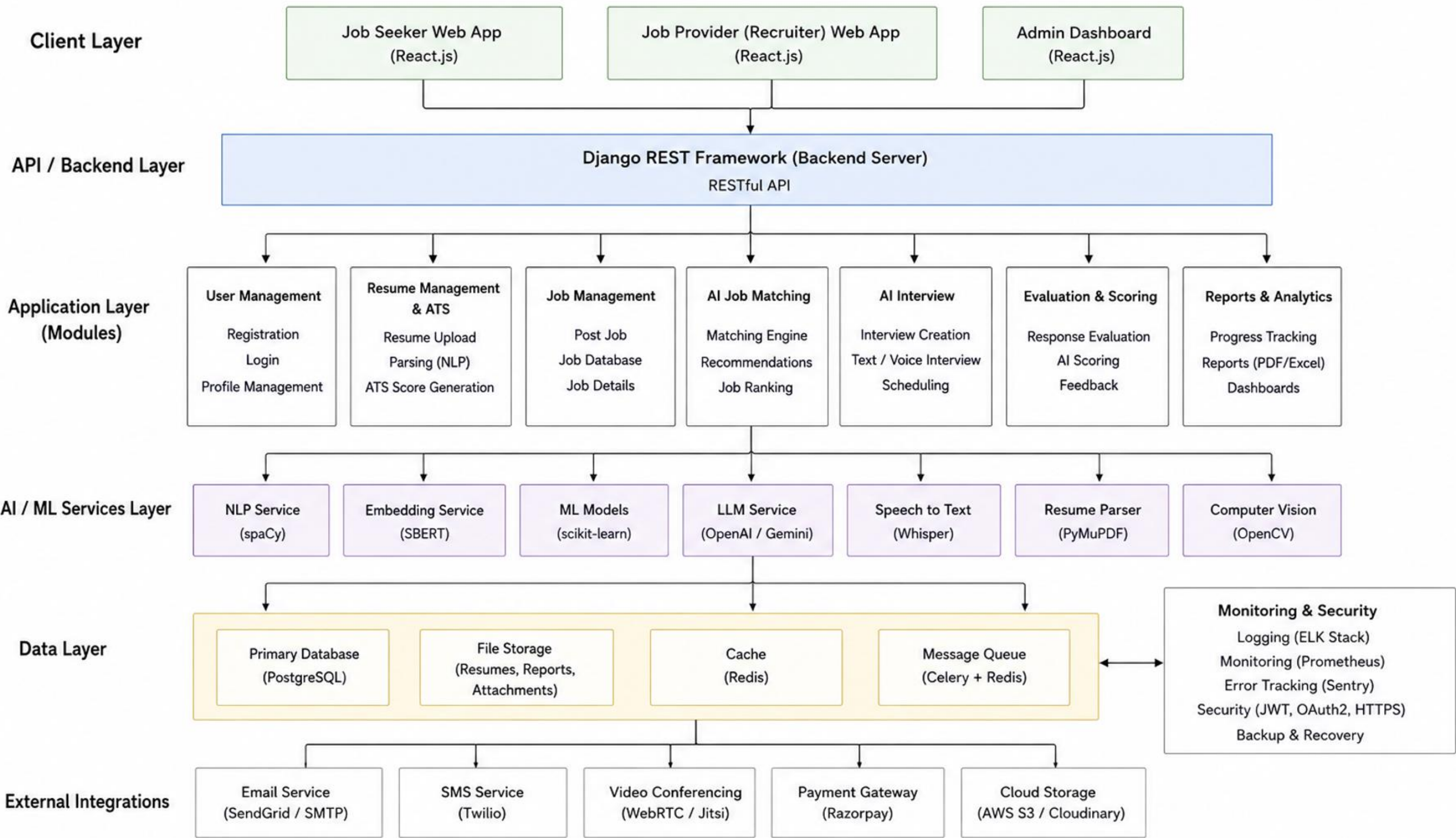
RESEARCH GAP

- Existing platforms specialize in individual stages of the recruitment process.
- Candidates must switch between multiple applications for resume improvement, interview preparation, and job search.
- Recruiters also rely on separate tools for candidate evaluation and hiring insights.
- JobRix unifies these capabilities into a single platform, supporting career readiness, AI-assisted assessment, job matching, and recruiter analytics through one integrated workflow.

Proposed Methodology - Technical Steps



JOBRIX – SYSTEM ARCHITECTURE



</> Technology Stack



Frontend

- React.js
- HTML5
- CSS3
- JavaScript (ES6+)
- Tailwind CSS
- Bootstrap



Backend

- Python
- Django
- Django REST Framework (DRF)
- RESTful APIs



Database

- PostgreSQL



AI/ML & Specialized Tools



OpenAI GPT-4o / Gemini API / Llama 3.1 (Ollama)

AI interview, evaluation, feedback & career guidance



OpenAI Whisper

Speech-to-Text conversion



OpenCV

Interview monitoring, face detection



spaCy

NLP for resume parsing, text processing



Sentence Transformers

Semantic similarity for job matching & recommendations



scikit-learn

Machine learning models for recommendations & scoring



PyMuPDF (fitz)

Extract text from PDF resumes



Pandas

Data manipulation & preprocessing



NumPy

Numerical operations & data handling



Development Tools



Visual Studio Code



Git



GitHub



Postman



Hardware / Environment Requirements

Hardware

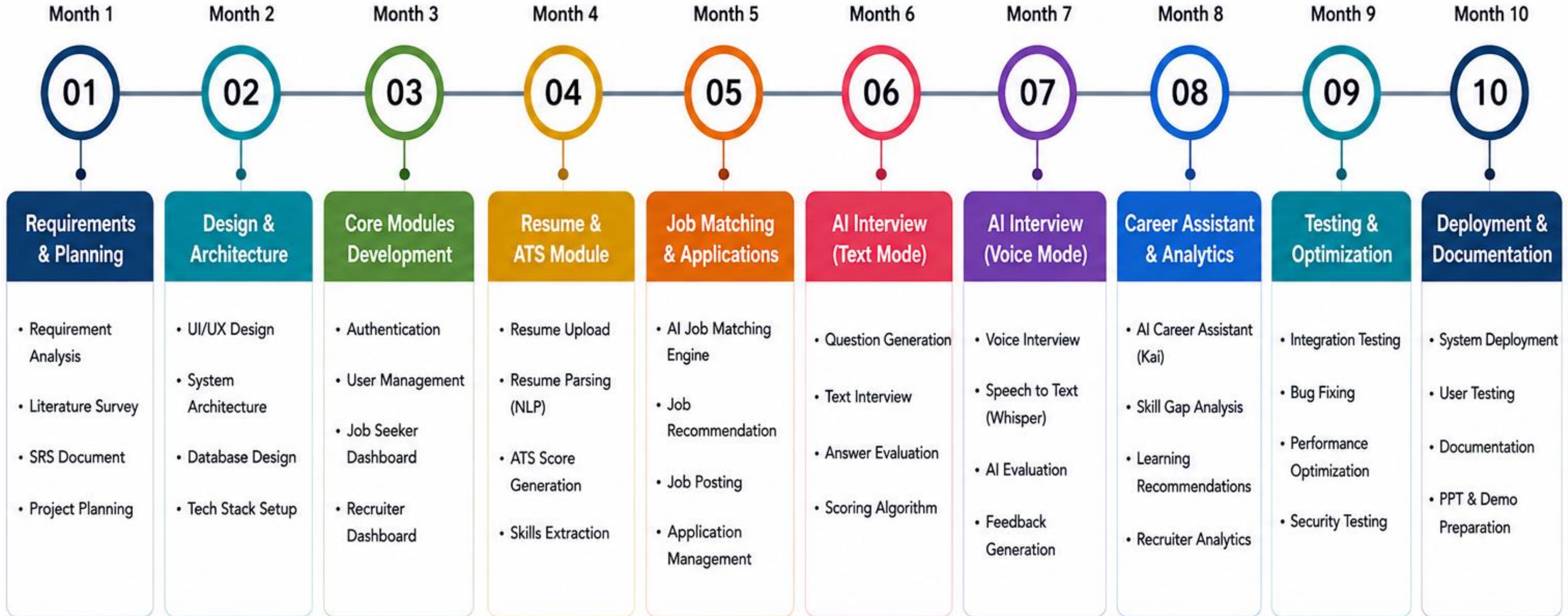
- Intel Core i5 / AMD Ryzen 5 or higher
- 8 GB RAM (16 GB Recommended)
- 256 GB SSD
- Webcam & Microphone
- Stable Internet Connection

Software Environment

- Windows 11 / Ubuntu 22.04 LTS
- Python 3.11+
- Node.js (v20+)
- PostgreSQL
- Git

JOBRIX – AGILE PROJECT TIMELINE (10 MONTHS)

—————> Iterative Development in Sprints (4 Weeks Each) <—————



AGILE PRINCIPLES FOLLOWED

Iterative
Development

Early & Continuous
Delivery

Regular Testing
& Feedback

Adaptability
to Changes

Collaboration &
Transparency



Expected Outcomes



Project Deliverables

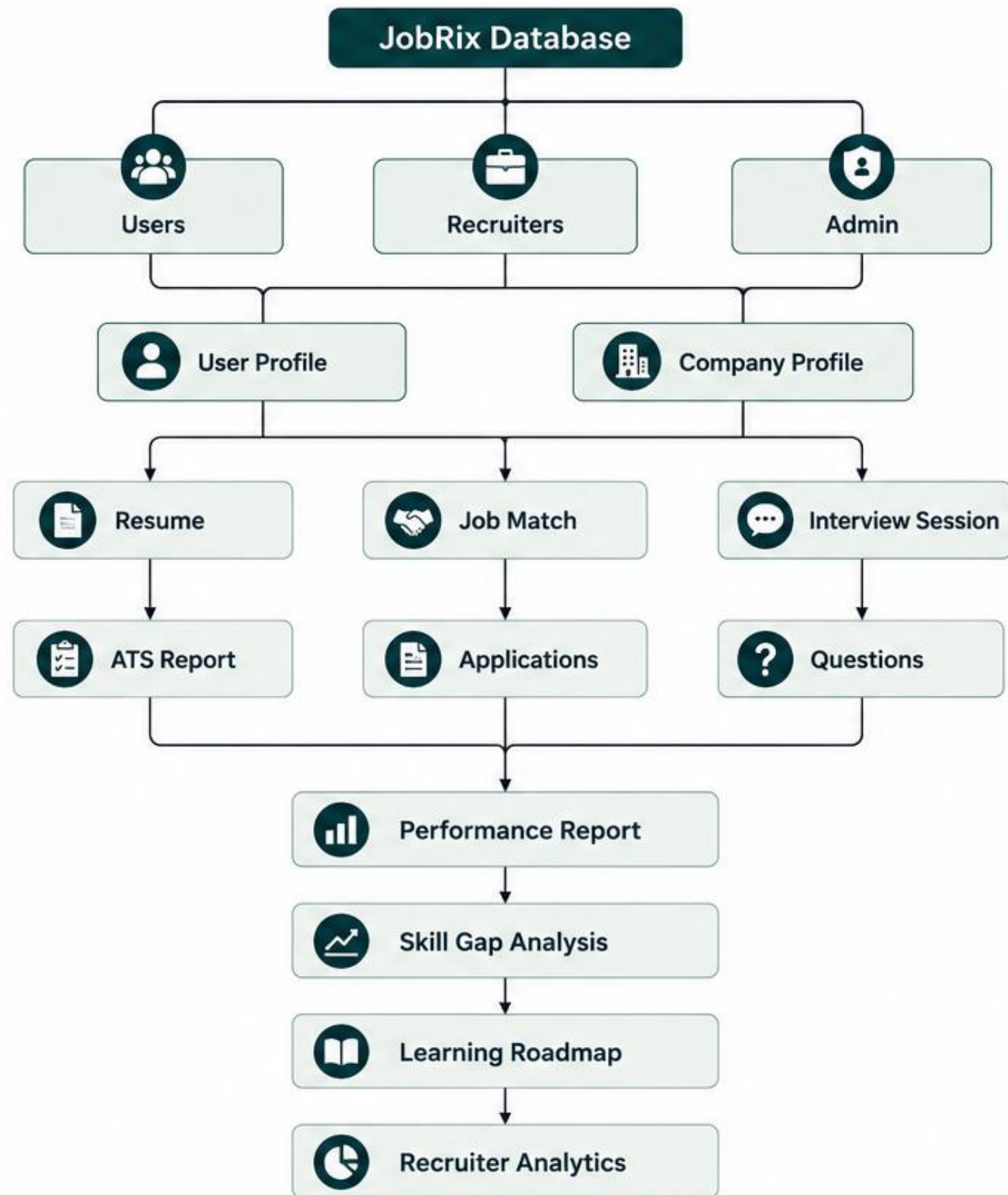
- 1 Develop an AI-powered recruitment platform with dedicated Job Seeker and Recruiter portals.
- 2 Implement secure authentication, profile management, ATS-based resume analysis, and AI-powered job recommendations.
- 3 Build AI-driven mock interview and Career Assistant modules for interview preparation and skill development.
- 4 Provide performance dashboards with interview reports, skill-gap analysis, progress tracking, and recruiter analytics.
- 5 Deliver a responsive web application with complete system architecture and technical documentation.



Future Scope

- 1 Integrate with major job portals (LinkedIn, Naukri, Indeed, Internshala) and enterprise HRMS/ATS systems.
- 2 Introduce AI-powered resume builder, company-specific interview simulations, and personalized learning roadmaps.
- 3 Expand to Android and iOS applications with multilingual support for global users.
- 4 Incorporate advanced AI features such as facial expression analysis, emotion recognition, eye-gaze tracking, and real-time coding assessments.
- 5 Add predictive hiring analytics, salary forecasting, market trend analysis, and blockchain-based credential verification.

Database Schema & Models



Core Database Models & Relationships

Core Database Models

1. Authentication

- User
- Recruiter
- Admin

2. Candidate

- UserProfile
- Resume
- ATSReport
- JobApplication

3. Interview

- InterviewSession
- InterviewQuestion
- InterviewResponse
- PerformanceReport

4. Recruitment

- Company
- JobPosting
- CandidateEvaluation
- RecruiterAnalytics

5. Career Development

- SkillGap
- LearningRecommendation
- ProgressTracker

Relationships

- One User → One Profile
- One User → Multiple Resumes
- One User → Multiple Job Applications
- One Job → Multiple Applicants
- One Interview → Multiple Responses
- One User → Multiple Performance Reports



Reference Sources

(Links for Web/App Based Projects)

1



React Documentation

<https://react.dev/>

2



Django Documentation

<https://docs.djangoproject.com/>

3

DRF

Django REST Framework

<https://www.django-rest-framework.org/>

4



PostgreSQL Documentation

<https://www.postgresql.org/docs/>

5



OpenCV Documentation

<https://docs.opencv.org/>

6



OpenAI API Documentation

<https://platform.openai.com/docs>

7



OpenAI Whisper Documentation

<https://platform.openai.com/docs/guides/speech-to-text>

8



spaCy Documentation

<https://scikit-learn.org/stable/>

9



Scikit-learn Documentation

<https://www.sbert.net/>

10

SBERT

Sentence Transformers Documentation

<https://www.sbert.net/>

11



PyMuPDF Documentation

<https://pymupdf.readthedocs.io/>

12



GitHub

<https://github.com/>



Reference Sources

(Links for Research paper referred)

1

Stanford HAI Study on Commercial AI Screening Vendors

<https://news.stanford.edu/stories/2026/06/ai-hiring-tools-racial-bias-research>

2

Benchmark Study on BERT-Based Candidate Ranking Systems

[https://www.researchgate.net/publication/406999123 Automated Resume Screening and Candidate Ranking System Using BERT-based Contextual Embeddings](https://www.researchgate.net/publication/406999123_Automated_Resume_Screening_and_Candidate_Ranking_System_Using_BERT-based_Contextual_Embeddings)

3

Study on Legacy ATS Platforms (Workday, Taleo, Greenhouse)

<https://www.managementpaper.net/archives/2025/vol7issue2/PartG/7-2-132-845.pdf>

4

Research on HireVue & Video Interview Platforms

[https://www.researchgate.net/publication/380357056 AI-BASED RECRUITING PLATFORM IN THE RECRUITMENT PROCESS AND QUALITY OF HIRING](https://www.researchgate.net/publication/380357056_AI-BASED_RECRUITING_PLATFORM_IN_THE_RECRUITMENT_PROCESS_AND_QUALITY_OF_HIRING)

Any Questions??